

PAVAN VIGNESH TIRUPATHI

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PROFILE

- CCNA-certified Network engineer with expertise in troubleshooting, routing, switching, and IP services.
- Currently at Cisco, supporting enterprise-scale networks with technologies such as OSPF, BGP, VLAN, DHCP, DNS and more.
- Skilled in Cisco IOS, Wireshark, and Linux tools for diagnosing and resolving real-time network issues.
- Familiar with Python for automation and basic scripting to streamline network operations.
- Strong foundation in designing, testing, and supporting reliable and secure network solutions.

TECHNICAL SKILLS

- **Networking:** TCP/IP, routing protocols (RIP, EIGRP, OSPF), switching protocols (VLANs, STP, VTP), and network security protocols (ACLs, DHCP snooping, DAI, port security), Network Troubleshooting, Bug Scrubbing, IP Services (DNS, DHCP), Network Security, Cisco IOS.
- **Network Devices:** Cisco ISR 4000 Series Routers, Cisco ASR 1000 & 9000 Series Routers, Cisco Catalyst 2000, 3000 & 9000 Series Switches
- **Programming Languages:** Python(NumPy, Pandas), C, C++
- **IoT & Embedded:** ESP8266/ESP32, Arduino, MQTT, HTTP, REST APIs, ThingSpeak.
- **Tools & Operating Systems:** Git, Wireshark, Linux & Windows

EXPERIENCE

Consulting Engineer Trainee Cisco Systems, Inc

Bengaluru Karnataka
September 2024 – Present

- Developed and implemented a system for efficient bug tracking, improving bug identification and categorization for quicker resolution.
- Systematically analyzed reported bugs, categorized them according to severity, and worked closely with development teams to prioritize resolutions.
- Conducted proactive software recommendations for different software trains for large-scale networks.
- Work with developers to address and fix the bugs, collaborating as needed to test solutions and verify that the issue has been resolved.
- Facilitated communication between QA and development teams, ensuring critical bugs were swiftly addressed, boosting team productivity.

Operations & Strategy Associate Appleton Innovations

Visakhapatnam, Andhra Pradesh
November 2023 – June 2024

- Optimized internal workflows across product development, delivery, and support.
- Designed and delivered IoT training workshops for 400+ students across 4 engineering colleges.
- Managed client communications and worked with different teams to keep projects on track.
- Created pitch decks, business plans, and product content for incubators and competitions.
- Achieved selection for the prestigious OCP 3.0 incubation program at Kalpataru CoE, STPI, for an Industrial IoT project, chosen from 30 competing teams.

CERTIFICATION

Cisco Certified Network Associate (CCNA 200-301)

Issued December 2024

WORK HIGHLIGHTS

Title: Tool for Network Configuration Analysis

Associated with : Cisco Systems, Inc

Description: The project aimed at automating the analysis of Cisco router and switch configuration files. Traditionally, engineers had to manually review large configs to identify enabled protocols and features. I contributed by developing Python code that parses IOS/IOS XE configs and generates structured reports highlighting active routing protocols (OSPF, BGP), services (SNMP, NTP, AAA), ACLs, VLANs, and interface details. This helped reduce manual effort and improved accuracy during configuration audits.

- Designed Python modules to parse configuration files and extract features using regex-based pattern matching.
- Implemented reporting in JSON format and supported integration for CLI/HTML output.
- Performed unit testing, debugging, and code reviews to ensure accuracy of results.
- Collaborated with the team in extending functionality for additional protocols and reporting requirements.

Title: Software Analysis and Bug Scrub Reporting for Enterprise Customers

Associated with : Cisco Systems, Inc

Description: As part of Cisco's CX team, I contributed to preparing Software Analysis and Release Standards reports for multiple multinational clients. These reports involved analyzing Cisco IOS-XE software releases for routers and switches, identifying potential software bugs, reviewing severity and risk levels, and providing recommendations tailored to the customer's environment. The reports helped customers plan safe upgrades, mitigate risks, and improve network stability.

- Analyzed defects from Cisco's internal bug databases and mapped them against customer requirements (hardware, software, features, performance, security).
- Prepared detailed bug scrub reports, highlighting severity-1/2/3 defects, risk impact summaries, and fixed release recommendations.
- Contributed to drafting and reviewing final SARS documents shared with customers.
- Collaborated with senior engineers to ensure alignment with business goals and operational readiness of enterprise networks.
- Supported post-report discussions with customers to clarify defect impact and upgrade paths.

EDUCATION

- Bachelor of Technology (Electronics & Communication Engineering) - GITAM University, CGPA - 8.33
- Intermediate(11th, 12th) - Board of Intermediate Education, Andhra Pradesh, CGPA - 9.34
- SSC(10th) - Board of Secondary Education, Andhra Pradesh, CGPA - 10

ACHIEVEMENTS

- Achieved selection for an Industrial IoT project in the prestigious Open Challenge Program 3.0 incubation by Software Technology Parks of India (STPI) at Kalpataru Center of Excellence, chosen among 30 competing teams.
- Took active participation in organizing and conducting sessions for multiple NGOs as a part of CSR activities